TONGMIAO XU

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RESEARCH INTERESTS

My research interests include enhancing robotic manipulation and interaction with the physical world through efficient control, estimation, and learning algorithms. My long-term goal is to utilize robotics to automate various tasks in daily life scenarios given human instructions, thereby seamlessly integrating into homes and factories to support humanity.

EDUCATION

MS in Computer Science Aug 2025 – May 2027(expected)

- University of Illinois Urbana-Champaign, GPA 4.0/4.0 Urbana, IL
  Related courses: Mobile Robotics(**SLAM, kalman filter..) A+**; Database System; Deep Generative Model; Reinforcement Learning; Deep Learning for Computer Vision

BS in Electrical Engineering, Mechanical Engineering (Double major) Sep 2021 – Jun 2025

- Xi'an Jiaotong University, GPA: 3.90/4.0 in Honors Engineering Program Xi'an, China
  Related courses: Program Design Method; Computer Network Theory; Data Science; Artificial Intelligence
- Study Abroad to **University of California, Berkeley** in Fall 2023 Berkeley, CA
  Related Courses: Mechatronics Design; Introduction to Embedded Systems; Advanced Control Systems

PUBLICATIONS

Dynamic Vision-Enabled Machine Condition Monitoring: A Point Cloud-Based Diagnostic Methodology.
T. Xu*, D. Cai, S. Yang, S. Gu, X. Li. PHM-Beijing 2024. PDF | Code

Sim-to-Real Image Translation for Real-to-Sim Policy Evaluation.
Tongmiao Xu, Shivansh Patel, Svetlana Lazebnik. **ICRA 2027** Under Review Website

RESEARCH EXPERIENCE

University of Illinois Champaign, IL
Research Assistant, ***Real-to-Sim*** *Robot Policy Evaluation (Advisor: Prof. Svetlana Lazebnik, Yunzhu Li)* Sep 2025 – Present

- Built a simulation **policy evaluation** pipeline using MuJoCo and 3D Gaussian Splatting for photorealistic reconstruction.
- Trained an image turbo model with DINOv3 features for end-to-end sim-to-real **image translation**, demonstrating that closing the visual domain gap improves correlation by nearly 30% over classical affine image transformation.
- Investigated **key design parameters** for policies including ACT, Diffusion Policy, and π0.5 VLA for faithful correlation.
- Developed real-to-sim **dynamics matching**, achieving close trajectory-level alignment between sim and real.

National University of Singapore Singapore
Research Intern in Robot Learning with Dr. David Hsu *academic poster* Jul 2024 – Apr 2025

- Developed **object-centric imitation learning models** for robotic manipulation (Fetch robot).
- Advanced **3D representation learning** with RLBench to improve generalization by 67.5% across views/objects.
- Presented work as an **academic poster at Eastern European Machine Learning Summer School (EEML 2025)**.

Xi'an Jiaotong University Xi'an, China
Research Assistant in Machining Learning with Dr. Xiang Li Nov 2022 – May 2024

- Applied **event cameras** and **PointNet++** for machine fault diagnosis with point cloud representations.
- Built novel **geometric data structure** for event streams; achieved robust classification under variables.
- Published at **2024 IEEE Global Reliability & PHM Conference**.

TEACHING EXPERIENCE

Siebel School of Computing & Data Science, University of Illinois Champaign, IL
Graduate Teaching Assistant for CS107: Data Science Discovery and Database System Aug 2025 – Dec 2025

- Lead weekly Python labs and office hours for machine learning, database system for up to 60 students.
- Collaborate with professors to design course materials and assignments, demonstrating strong communication and teamwork skills recognized by faculty.

Improve students' practical coding skills and contribute to curriculum quality and receive a full TA evaluation.

Siebel School of Computing & Data Science, University of Illinois Champaign, IL
Graduate Teaching Assistant for CS411: Database System Jan 2026 – May 2026

- Lead weekly office hours and Campuswire support for database systems topics including SQL, query optimization, and machine learning pipelines for up to 100 students.

- Collaborate with professors and TAs to design, test, and grade machine problems (MPs) and exams

Mentor students through hands-on programming assignments improving their coding skills in database implementation.

INDUSTRY EXPERIENCE

SceniX/WorldLabs New York, US

Robot Learning Intern May 2026 – Aug 2026

Project: Sim-to-Real Robot Manipulation

- Built an end-to-end sim-to-real pipeline for xArm singulation tasks (pencils, markers, bolts), sustaining 1-hour continuous autonomous runs with no human intervention at 159 pencils/hr, 145 markers/hr, 130 bolts/hr
- Trained manipulation policies entirely on simulated data and rendered images, with no real-world data, and transferred them to strong real-world performance.
- Developed the full pipeline stack: simulation data collection, MPPI-based recovery, Unreal Engine data rendering, policy training, and real-robot deployment.

Theia IoT Technology Co. Ltd. Xi'an, China

Software Development Department: Dec 2023 – Jan 2024

Project: Embedded Real-Time Mobile Object Detection

- Developed applications in a Qt C++ environment with OpenCV, deployed on Nvidia Jetson units under a Linux system.
- Deployed traditional detection algorithms, like Gaussian Mixture Mode, Optical Flow, and KCF, for real-time detection.

RELEVANT COURSEWORK&SCI-TECH COMPETITION PROJECT

"Minato Cup" 2024 China Undergraduate Mechanical Engineering Innovation Competition Fall 2024

Awarded the First Prize in the 2024 China College Students Mechanical Engineering Innovation and Creativity Competition

- Designed the PCB circuit board optimized for high-accuracy resistance measurement.
- Deployed an embedded application on a Raspberry Pi 3 Model B board, ensuring efficient functionality.

Mechanical Art Installation "Tears of the Ocean" with ESP32-based PID Control Fall 2023

ME102B course (Mechatronics Design) Project at UC Berkeley

Awarded the Second Prize in Mechatronics Design Social Impact at UC Berkeley

- Designed and machined the mechanical parts of the entire system, ensuring functionality and structural integrity.
- Implemented precise rotation angle control of the moving mechanisms, including a motor carrying 13 heavy wooden plates with a shaft, using PID control based on force-sensitive resistor data.

Raspberry Pi Based Bluetooth Rescue Rover Fall 2023

EECS149 course (Introduction to Embedded & Cyber-Physical Systems) Project at UC Berkeley

- Enabled motion control for a 3pi+ 2040 robot equipped with an HC05 module, operating through Bluetooth commands.
- Implemented a finite state machine and utilized gyroscopes to perform localization.

All-Terrain Adaptive Path Following and Obstacle Avoidance Robot Spring 2023

Awarded the Excellence Prize in the 25th Shaanxi Province China Robot Competition

- Designed a small robot vehicle tailored to the competition arena specifications, optimizing performance and adaptability.
- Integrated motor and servo actuation with state machine software, leveraging ultrasonic and color sensors to enable all-terrain adaptive path following, line tracking, obstacle avoidance, and projectile launch command execution.

SKILLS

Programming: Python, C/C++, MATLAB
Softwares & Tools: ROS, PyTorch, TensorFlow, OpenCV, Solidworks, Gazebo, Git, LaTeX
Hardware: Arduino, Raspberry Pi
OS: Linux

AWARDS & HONORS

- The International Journal of Structural Integrity Prize at the 2024 Global Reliability & Prognostics and Health Management Conference Oct 2024
- National University Mechanical Engineering Innovation and Creativity Competition 1st Prize Aug 2024
- 34th XJTU "Tengfei Cup" Sci-Tech Competition 2nd Prize Apr 2023
- National College Students Mathematics Competition 3rd Prize, Shaanxi Region Mar 2023
- Hsue-shen Tsien Scholarship, XJTU (top 5%) Dec 2022
- Outstanding Student Awarded by Hsue-shen Tsien Honors College (top 5%) Dec 2022
- Outstanding Individual by Hsue-shen Tsien Alumni Award Nov 2022
- National College Student Mathematical Modeling Competition 1st Prize, Shaanxi Region (top 10%) Nov 2022

- National University Intelligent Robot Contest 1st Prize, Shaanxi Region (top 10%) Aug 2022
- National 3D Digital Innovation Design Contest 1st Prize, Shaanxi Region (top 5%) Jul 2022

ACTIVITIES & LEADERSHIP

- Chair of the Class Cultural and Sports Committee Sep 2021 – Present
- Leader of the Hsue-shen Tsien Academic Study Assistance Group Aug 2022 – Aug 2023
- Excellent Volunteer of the "Caring for Shangluo" Project (selection rate: 10%) Jun 2022 – Jul 2022